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DEPARTMENT OF EDUCATION

COMPUTER SYSTEMS SERVICING NC II
GRADE 11

QUARTER 1
COC 1:
Install and Configure Computer Systems

Module 2:
Parts and Functions of Computers

Computer Systems Servicing – Grade 11
Quarter 1 – Module 2: Parts and Functions of Computer
First Edition, 2020

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Introductory Message

For the facilitator:

Welcome to the **Computer Systems Servicing Grade 11 Module on Parts and Functions of Computer**

This module was collaboratively designed, developed and reviewed by educators from Schools Division Office of Pasig City headed by its Officer-In-Charge Schools Division Superintendent, Ma. Evalou Concepcion A. Agustin in partnership with the Local Government of Pasig through its Mayor, Honorable Regis Maria Victor Sotto. The writers utilized the standards set by the K to 12 Curriculum using the Most Essential Learning Competencies (MELC) while overcoming their personal, social, and economic constraints in schooling.

This learning material hopes to engage the learners into guided and independent learning activities at their own pace and time. Further, this also aims to help learners acquire the needed 21st century skills especially the 5 Cs namely: Communication, Collaboration, Creativity, Critical Thinking and Character while taking into consideration their needs and circumstances.

In addition to the material in the main text, you will also see this box in the body of the module:

Notes to the Teacher

This contains helpful tips or strategies that will help you in guiding the learners.

As a facilitator you are expected to orient the learners on how to use this module. You also need to keep track of the learners' progress while allowing them to manage their own learning. Moreover, you are expected to encourage and assist the learners as they do the tasks included in the module.

For the learner:

Welcome to the Computer Systems Servicing Grade 11 Module on Parts and Functions of Computer

The hand is one of the most symbolized part of the human body. It is often used to depict skill, action and purpose. Through our hands we may learn, create and accomplish. Hence, the hand in this learning resource signifies that you as a learner is capable and empowered to successfully achieve the relevant competencies and skills at your own pace and time. Your academic success lies in your own hands!

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning material while being an active learner.

This module has the following parts and corresponding icons:

Expectation - These are what you will be able to know after completing the lessons in the module

Pre-test - This will measure your prior knowledge and the concepts to be mastered throughout the lesson.

Recap - This section will measure what learnings and skills that you understand from the previous lesson.

Lesson- This section will discuss the topic for this module.

Activities - This is a set of activities you will perform.

Wrap Up- This section summarizes the concepts and applications of the lessons.

Valuing-this part will check the integration of values in the learning competency.

Post-test - This will measure how much you have learned from the entire module.

EXPECTATION

After reading this INFORMATION SHEET, YOU MUST be able to:

1. identify different parts and functions of a computer;
2. explain the different parts and functions of a computer;
3. appreciate the importance of parts and functions of computer.

PRE-TEST

Direction: Identify what is referred to in the following statements. Write your answers on the lines.

_____1. It is also known as the "Brain of the computer".

_____2. Found on many laptops, this square pad lets you move the cursor by dragging your finger across its surface.

_____3. Used to interact with items on your computer screen.

_____4. This pointing device looks like a pencil eraser protruding from the middle of your keyboard.

_____5. Used to display text and images on the screen.

_____6. One of the main ways to communicate with a computer.

_____7. Allows you to play music and hear sounds effects from your computer.

_____8. These keys include the same letter, number, punctuation, and symbol found on a traditional typewriter.

_____9. A flat panel display, which uses an array of light-emitting diodes as a video display.

_____10. This keypad is handy for entering numbers quickly.

RECAP

Last session we tackle about what is computer the different types of computers, and its different sizes and power.

A computer is a machine programmable to manipulate the symbols. Its main characteristics are:

- It gives a well-defined response to a particular set of instructions.
- This will execute a pre-programmed list of instructions.
- It can store and recover vast volumes of data easily.

Types of Computers

- **Personal computer:** A small, single-user, microprocessor-based computer.
- **Workstation:** Single-user, strong computer. A workstation is like a personal computer, but it does have a more powerful microprocessor, and a higher quality display in general.
- **Minicomputer:** A multi-user computer which can accommodate up to hundreds of users at the same time.
- **Mainframe:** A powerful multi-user computer capable of supporting many hundreds or thousands of users simultaneously.
- **Supercomputer:** An extremely fast computer that can perform hundreds of millions of instructions per second.

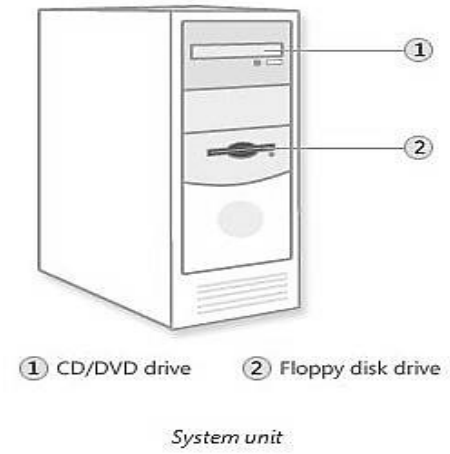
LESSON

BASIC PARTS OF COMPUTER

1. System Unit

The system unit is the core of a computer system. Usually it's a rectangular box placed on or underneath your desk. Inside this box are may electronic components that process information. The most important of this component is random access memory (RAM), which temporarily stores information that the CPU uses while the computer is on. The information stored in RAM is erased when the computer is turned off.

Image 1.1
System Unit

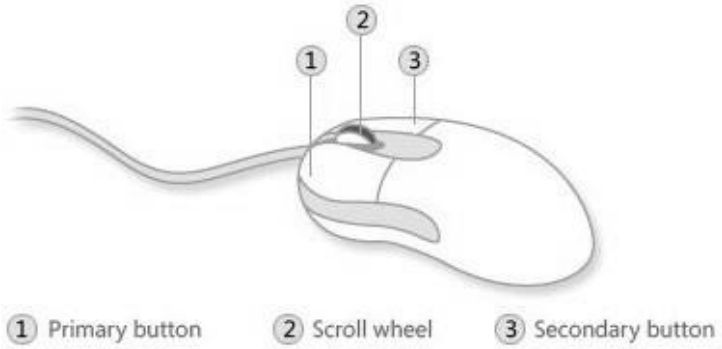


Almost every part of your computer connects to the control panel by using cables. The cables are plugged into specific ports (openings), typically at the back of the system unit. Hardware that is not part of a network configuration is often referred to as a peripheral computer or device.

2. The Mouse

The mouse is used to communicate with the objects on the computer screen. You can move objects, open them, change them, throw them away, and perform other actions, all by pointing and clicking your mouse.

Image 1.2 *The mouse*



Kinds and parts of Mouse

- **Mouse ball:** A little rubber ball lies in a mouse 's belly; you roll the little ball, too, as you push the mouse. Moving the ball tells the computer the direction and speed to move the on-screen pointer.
- **Optical:** Optical mice ditch the ball/roller mechanics for a small glowing light and a sensor.
- **Trackball:** Trackballs are, in essence, upside-down mice.
- **TrackPoint/AccuPoint:** Found on some laptops, this pointing device looks like a pencil eraser protruding from the middle of your keyboard.
- **Touchpads:** Found on many laptops, this square pad lets you move the cursor by dragging your finger across its surface.
- **Wireless:** Wireless mice work just like their keyboard counterparts; in fact, some share the same receiver unit that plugs into the USB or PS/2 port of your computer.

PS/2: An older mouse comes with a connector in the PS/2 style which still works well. Just don't unplug the mouse while the computer is turned on, the mouse will stop working even after you plug it back in frenziedly. (Restart the computer and the mouse will start working again.)

3. The Monitor

Video circuits on your PC send images to your monitor, where you can see the action. Because monitors and video circuits on your PC (known as video cards or display adapters) work together as a team. These terms appear on newspaper advertisements, showroom signs and the fine print of product boxes while shopping for either a computer or video card.

Monitors come in different types, each described below.

- **CRT (cathode ray tube)**

By comparison, old school CRT monitors seem boring and bulky.

Fading quickly from the marketplace, shown in Figure 3-2, CRT (Cathode Ray Tube) monitors mimic small (but expensive) television sets. Although some CRT monitors call themselves "flat screen," their glass screens are relatively flat, that means only. They are not flat panel monitors, an honor which only belongs to LCD monitors.

Image 1.3
CRT (cathode ray tube)

LCD (Liquid Crystal Display)

LCD Monitors look slim and hip on any desktop. The most popular monitor today, LCD monitors look much like large laptop screens mounted on a stand. LCD monitors are also called flat-panel monitors.

Image 1.4
LCD (liquid Crystal Display)

- **Quantum dot LED**

A display technology that emits light using quantum dots. QLED is expected to be more effective than OLED, and to be less costly to manufacture. Also, QLED is ultrafine, translucent and versatile.

- **OLED (organic light emitting diode)**
  A display technology that provides bright, colorful images for sports and action movies with a wide viewing angle, low power, high contrast ratio and fast response time. In plasma and LCD / LED monitor / display the OLED technology varies greatly from the displays.

- **LED (light emitting diode)**
  An LED display is a flat panel display that uses a range of light-emitting diodes to display videos. An LED panel is a small display, or the larger display component.

Image 1.5
LED (light emitting Diode)

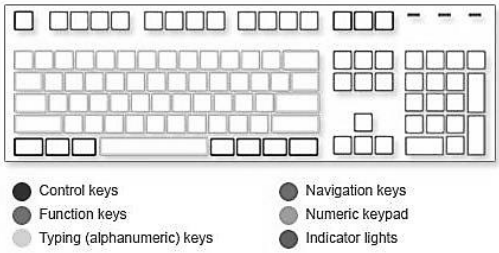
4. Keyboard

Keyboard is the main way to enter information into your computer. But did you know you can also control your computer using your keyboard? Learning just a few simple keyboard commands (computer instructions) can help you to work more effectively.

The keys on your keyboard can be divided into several groups based on function:

- **Typing (alphanumeric) keys.** These keys include the same letter, number, punctuation, and symbol keys found on a traditional typewriter.
- **Control keys.** These keys are used alone or in combination with other keys to perform certain actions. The most frequently used control keys are Ctrl, Alt, the Windows logo key and ESC
- **Function keys.** The function keys are used to perform specific tasks. They are labeled as F1, F2, F3, and so on, up to F12. The functionality of these keys differs from program to program.

Image 1.6
keyboard



How the keys are arranged on a keyboard

- **Navigation keys.** These keys are used for moving around in documents or webpages and editing text. They include the arrow keys, Home, End, Page Up, Page Down, Delete, and Insert.
- **Numeric keypad.** The numeric keypad is handy for entering numbers quickly. The keys are grouped together in a block like a conventional calculator or adding machine.

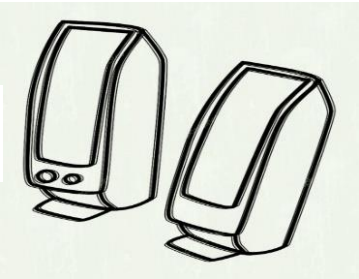
Additional Information:

- *Keyboard comes with three ports supported technology USB, PS2, and Wireless.*
- *Specialized keyboard keys require special drivers. Those specialized keys won't work until you install the keyboard's bundled software.*
- *Wireless keyboards bear no cords, making for tidy desktops. Most come in two parts: the keyboard and a receiving unit, which plugs into your PC's USB part. Unfortunately, they're battery hogs.*

5. Speaker

Speakers are used to play sound. They can be built into the system unit or connected with cables. Speakers allow you to listen to music and hear sound effects from your computer.

Image 1.7
Speaker



6. Printer

While it is not mandatory for a computer's overall operation, it is a useful tool which helps document management. It allows a user to print objects, such as letters and photographs, on paper.

Like most computer peripheral, printers come with their own secret vocabulary.

Image 1.8 *from left to right, Inkjet Printer, Laser Printer, All in One Printer Image*

Kinds of Printer

a. Inkjet

Popular for their low price and high quality, inkjet printers squirt ink onto a page, creating surprisingly realistic images in color or black and white.

b. Laser

Laser printers might sound dangerous, but these printers) use technology similar to their ho-hum counterpart, copying machines; with toner, they scan images in the paper. Black and white laser printers cost a little more than inkjet printers, twice the price for laser printers in color. Though laser printers are unable to print digital photos, they are cheaper for general office paperwork in the long run.

c. **All-in-one (AIO)**

This type of printer is popular with small offices and incorporates a laser or inkjet printer, copying machine, scanner, and fax machine into one compact package.

ACTIVITIES

Direction: Identify the following pictures and write your answers on the line.

_____1.

4. _____

_____2.

5. _____

_____3.

WRAP – UP

Computer users need different types of computer systems. What the user does with the computer dictates the components and peripherals needed. Looking at the computer systems by purpose is a good place to start with design. The Basic parts of computer is the System Unit, Monitor, Mouse, and Keyboard also included are the peripheral devices such as speaker, and printer although it's not mandatory for a computer's overall operation.

VALUING

DIRECTION: Read and answer the following questions carefully in two to three sentences each number.

1. What do you think are the importance of learning the parts and function of a computer?

   _____
   _____
   _____
   _____
   _____
   _____
   _____
   _____

2. How will you use the knowledge you acquired about parts and function of a computer?

   _____
   _____
   _____
   _____
   _____
   _____
   _____

POST – TEST

Direction: Write *T* on the line if the statement is correct and *F* if the statement is wrong.

__1. System unit is the core of a computer system.

__2. Almost every other part of your computer connects to the system unit using cables.

__3. When mice burst into the PC world in the early ’80s, Macintosh models had two buttons.

__4. Monitors and your PC’s video circuits (known as video cards or display adapter) work as one.

__5. All in-one printers quirt ink onto a page, creating surprisingly realistic images in color or black and white.

__6. Keyboard is an input device, the main way to enter information into your computer.

__7. Navigation keys are used for moving around in documents or webpages and editing text.

__8. Inkjet printers is popular for their high price and high quality.

__9. An older mouse comes with a PS/2 style connector.

__10. LCD monitors are also called flat-panel monitors.

KEY TOCORRECTION

Post- Test Key to correction

1. T
2. T
3. F
4. F
5. F
6. T
7. T
8. F
9. T
10. T

Pre- Test Key to correction

1. System Unit
2. Touch Pad
3. Mouse
4. Track Point
5. Monitor
6. Keyboard
7. Speaker
8. Typing(alphanumeric) keys
9. Light emitting Diode (LED) display
10. Numeric Keypad

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